



Scale SEAL Leaderboards: Frontier Model Performance Across 20+ Benchmarks

Comparative analysis of 100+ models across agentic coding, reasoning, safety, and multimodal benchmarks

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

< / > Agentic

- SWE Atlas
- SWE-Bench Pro
- MCP Atlas

○ Frontier

- Humanity's Last Exam
- EnigmaEval
- MultiChallenge
- SciPredict

⚠ Safety

- MASK
- PropensityBench
- Fortress
- Professional Reasoning

👁 Multimodal

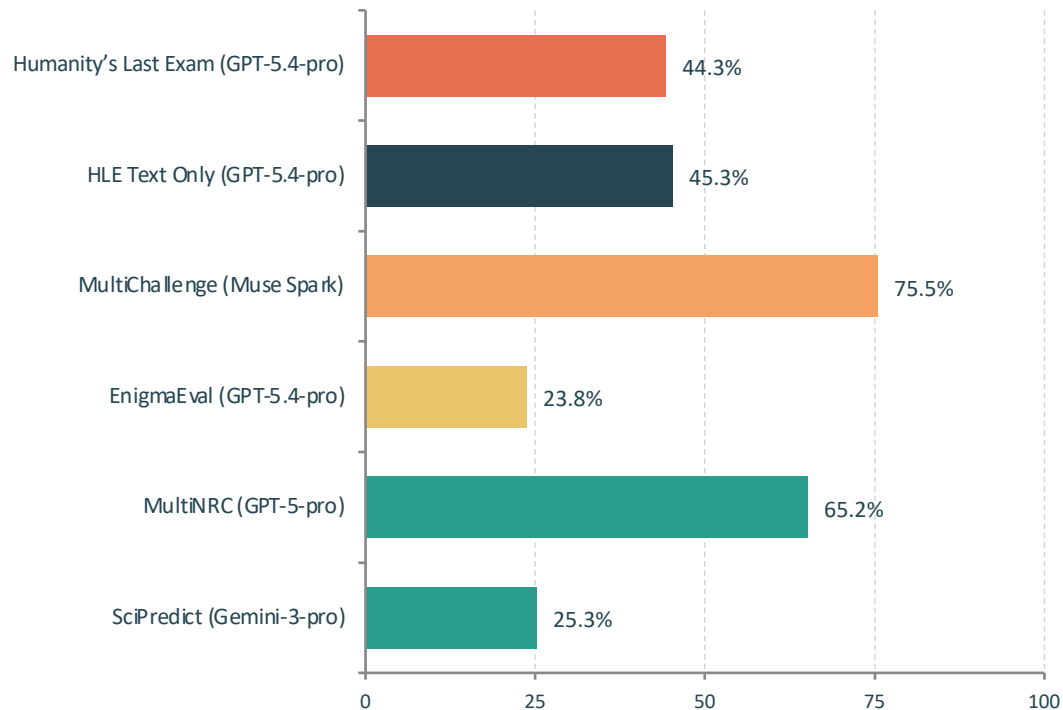
- AudioMultiChallenge
- VISTA
- VisualToolBench

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro

Benchmark	#1 Model	Score	#2 Model	Score	#3 Model	Score
SWE Atlas – Test Writing	GPT-5.4-xHigh (Codex CLI)	44.36	GPT-5.4-xHigh (Mini-SWE)	40.00	GPT-5.3-Codex-Xhigh	38.98
SWE Atlas – Codebase QnA	GPT 5.4 xHigh (Codex)	40.80	GPT 5.4 xHigh (Mini-SWE)	36.30	Opus 4.6 (Claude Code)	33.30
MCP Atlas (Tool Use)	Muse Spark	78.30	Claude Opus 4.6	75.80	Gemini 3.1 Pro	73.90
SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh)	59.10	Muse Spark	55.00	Claude Opus 4.6	51.90
SWE-Bench Pro (Private)	Claude Opus 4.6	47.10	Muse Spark	44.70	GPT-5.4-pro (xHigh)	43.40

Claude Opus 4.6 leads on private repositories (SWE-Bench Pro Private), while Muse Spark emerges as a strong new competitor across multiple benchmarks.

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%



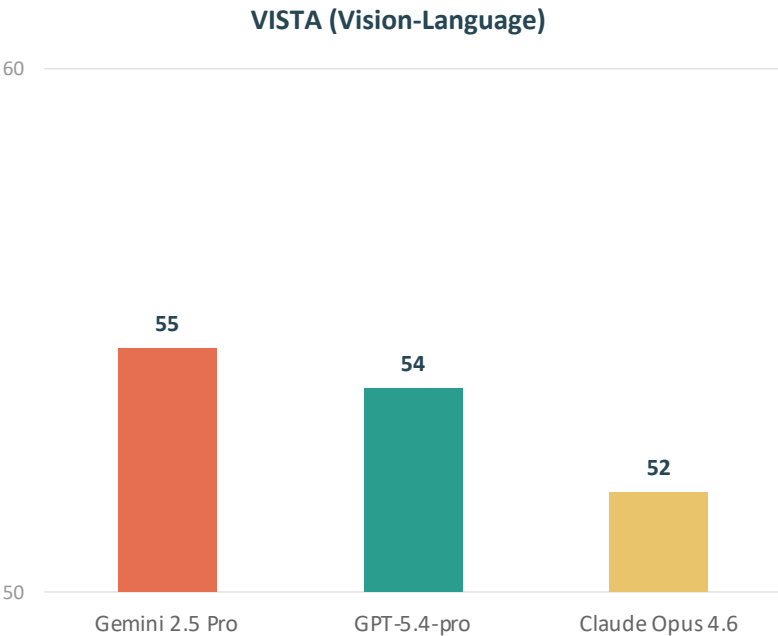
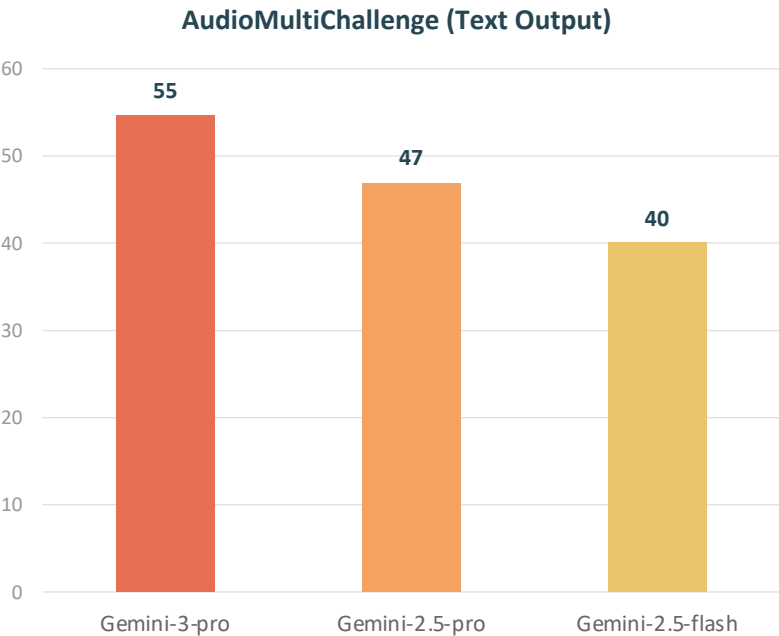
KEY INSIGHT

Even top scores remain well below saturation:

- HLE: 44% — far from ceiling
- EnigmaEval: 24% — complex multi-step reasoning
- SciPredict: 25% — scientific forecasting

These benchmarks are specifically designed to resist memorization and test genuine reasoning capability. Scores below 50% suggest frontier models still struggle with expert-level multi-domain problems.

Gemini Models Sweep Audio and Vision-Language Benchmarks



Audio Output: Gemini-3.1-flash-live leads at 36.06 vs GPT-realtime-1.5 at 34.73 | VisualToolBench: GPT-5.4 edges Gemini-3.1-pro (29.17 vs 28.97)

Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

MASK (Honesty)

Near Ceiling

Claude Opus 4.6

96.28%

Claude Sonnet 4

96.13%

~96%

Fortress (National Security)

Very Low

Top Score

8.24 ± 1.93

Claude Sonnet 4

12.80 ± 2.36

8–13

PropensityBench (lower = better): o3 at 10.50 — measures propensity to produce harmful content; lower scores indicate safer behavior

Safety alignment varies dramatically by task type — near-ceiling honesty vs. critically low national security scores

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Domain	Benchmark	#1 Model	Lab	Score
AGENTIC	SWE Atlas – Test Writing	GPT-5.4-xHigh	OpenAI	44.36
	SWE Atlas – Codebase QnA	GPT 5.4 xHigh	OpenAI	40.80
	MCP Atlas	Muse Spark	Muse	78.30
	SWE-Bench Pro (Public)	GPT-5.4-pro	OpenAI	59.10
	SWE-Bench Pro (Private)	Claude Opus 4.6	Anthropic	47.10
FRONTIER	Humanity's Last Exam	GPT-5.4-pro	OpenAI	44.32
	EnigmaEval	GPT-5.4-pro	OpenAI	23.82
	MultiChallenge	Muse Spark	Muse	75.52
	SciPredict	Gemini-3-pro	Google	25.27
SAFETY	MASK (Honesty)	Claude Opus 4.6	Anthropic	96.28
	Prof. Reasoning – Legal	Claude Opus 4.6	Anthropic	52.27
	Prof. Reasoning – Finance	Claude Opus 4.6	Anthropic	53.28
MULTIMODAL	AudioMultiChallenge (Text)	Gemini-3-pro	Google	54.65
	AudioMultiChallenge (Audio)	Gemini-3.1-flash	Google	36.06
	VISTA	Gemini 2.5 Pro	Google	54.65



Key Takeaways: No Single Model Wins Everything — Specialization Is the New Frontier

- 1 GPT-5.4 leads agentic coding and frontier reasoning — but not multimodal or safety
- 2 Gemini dominates audio and vision-language benchmarks, sweeping all top positions on AudioMultiChallenge
- 3 Claude leads safety and professional reasoning — 96%+ MASK honesty, #1 on Legal and Finance
- 4 Key benchmarks remain far from saturation: HLE (44%), EnigmaEval (24%), SciPredict (25%)
- 5 Muse Spark is a strong new competitor — top 3 on 6+ benchmarks, including #1 on MultiChallenge at 75.5%